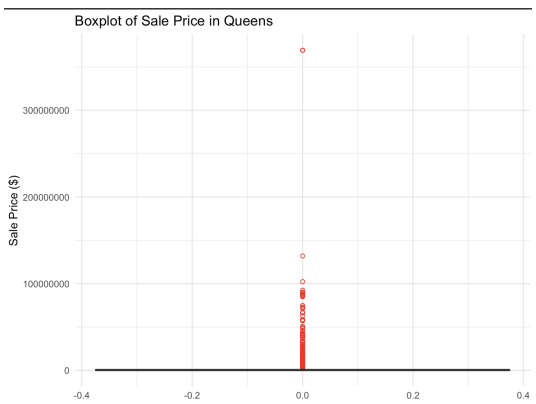
1. Regression

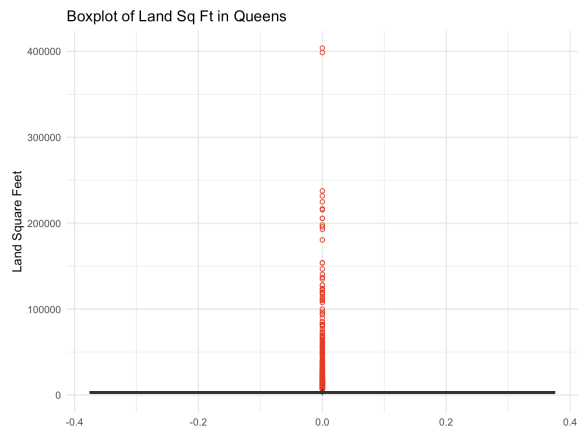
a)

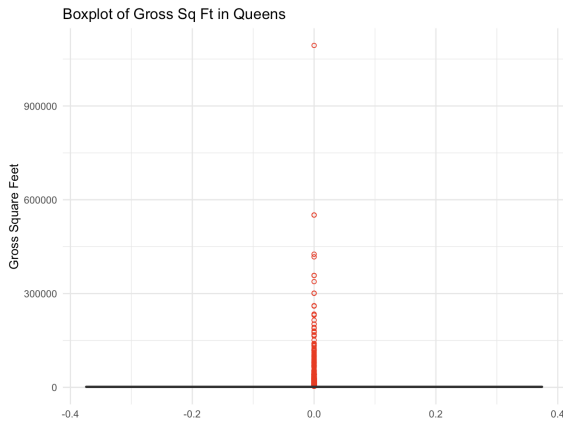
I created a subset containing data points from Queens only. Out of all the available features, I selected the following four variables to use in modeling: **Sale Price, Gross Square Feet, Land Square Feet, and Year Built.**

As part of the data cleaning process, I removed dollar signs and commas, and ensured these features were treated as numeric types. I then dropped all rows containing null values across the four features.

I generated initial boxplots to observe the distributions and identify outliers:





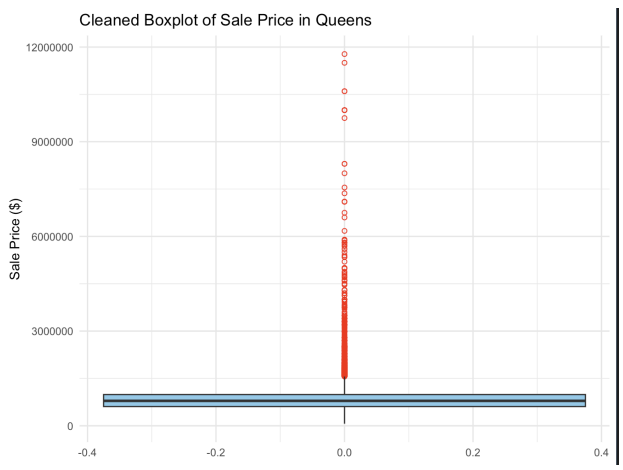


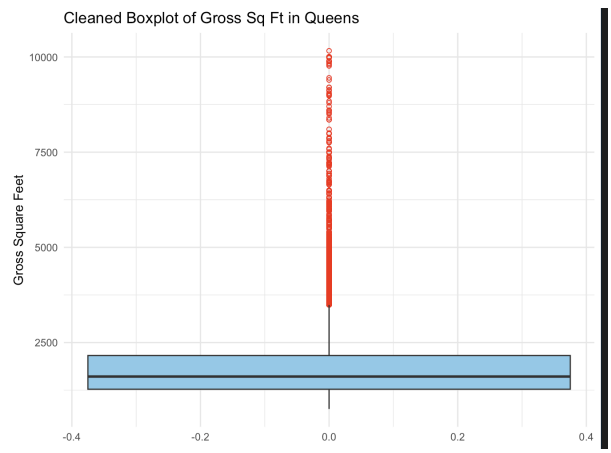
As we can see, the data was extremely right-skewed, we can barely see the boxplot.

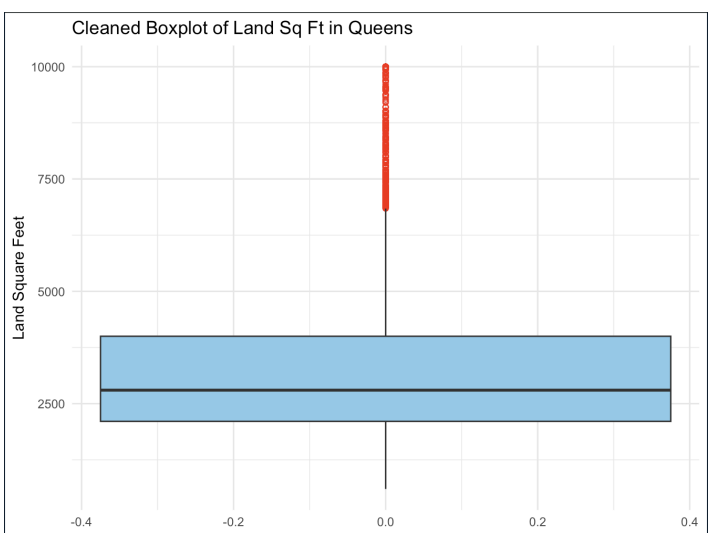
To handle the extreme outliers/bad data:

- **Sale Price:** I found there are about 15600 properties that have a sale price lower than 1000, which clearly correspond to non-market transactions. I removed them and then filtered the dataset to only include values within the 1st and 99th percentile.
- **Gross Square Feet:** I found there are about 1800 properties that are smaller than 100 sq ft, which is most likely an error. I removed them then applied the same percentile filtering.
- **Land Square Feet:** I only trimmed the top 1%. I did not trim the bottom percentiles because some properties in NYC probably don't have any land square footage.
- **Year Built:** I checked the minimum values and found no irregularities

After all of that, the boxplots became more accurate:







Note: I chose not to apply a log transformation to preserve the linear real world interpretability of the regression coefficients.

b)
I evaluated the models after I cleaned the data as I explained above.
I split the data into an 80% training set and a 20% testing set. I tested 3 different linear regression models with different feature combinations to predict the **Sale Price**: Model 1 (Gross Sq Ft), Model 2 (Gross Sq Ft + Land Sq Ft), and Model 3 (Gross Sq Ft + Land Sq Ft + Year Built).

The best model was the third one. It had the highest adjusted R-squared value (0.3687) which means it explains about 36.8% of the variance in the sale price. Plus, when it was evaluated against the testing data, it had the lowest root mean squared error (RMSE = 480,586.8) compared to Model 1 (482,432) and Model 2 (480,679.1). Finally, the summary output confirmed that all three features were statistically significant ($p < 0.001$).

2. Classification

a)

I chose the following 3 neighborhoods in Queens: Astoria, Long Island City, and Forest Hills. I kept the same outlier removals from part 1 and added 1 new step: I standardized all 4 quantitative features because for algorithms that rely heavily on distance calculations such as kNN and SVM, we don't want variables with bigger scales (e.g. Sale Price) to completely overpower smaller ones (e.g. year built).

Using a single 80/20 train/test split, I trained 3 supervised learning models:

- kNN, using a loop to choose the best k based on overall accuracy
- Radial SVM, using grid search to tune cost and gamma parameters
- Random Forest classifier (ntree=100)

The 4 features that were used to predict the neighborhood across all 3 models were: **Sale Price, Gross Square Feet, Land Square Feet, and Year Built.**

I evaluated the performances with contingency tables and the following metrics: precision, recall and F.

The results revealed that the **Random Forest** was the best model. It had the highest F1 score across the 3 classes: (Astoria: 0.82, Forest Hills: 0.71, Long Island City: 0.42). One thing to note is that all 3 models struggled with classifying properties from Long Island City. More on that in the Conclusions section.

3. Conclusions

1- The dataset is very messy and requires a lot of cleaning. It is heavily right-skewed and requires a lot of domain knowledge to be able to extract a correct representation of the housing market.

2- The regression model that was built using Gross Square Feet, Land Square Feet, and Year Built is unfortunately not complex enough to predict a property's exact value. However, the models worked well as a baseline.

3- Random Forest was the best model for predicting neighborhoods, which proves once again tree-based algorithms handle non-linear boundaries very well.

4- All classification models failed to accurately identify properties in Long Island City. For example, the Random Forest model misclassified 14 out of 23 actual Long Island City properties as Astoria (which is an adjacent neighborhood). There is also a severe class imbalance in the dataset: because the model had much less training examples for LIC (77) compared to Astoria (433), it became biased toward predicting the majority class.